Joel Rajah

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https://joelrajah08.github.io/

EDUCATION

New York University Tandon School of Engineering **New York, NY**
B.S. Computer Engineering Expected Graduation 2029
Relevant Coursework: Data Structures & Algorithms, Fundamentals of Electric Circuits, Linear Algebra & Differential Equations

TECHNICAL PROJECTS

AI Face Recognition App | ***(Python, Qt, NumPy, OpenCV, MediaPipe, dlib)*** (January 2026 - March 2026)
- Built a real-time face recognition application utilizing OpenCV for image processing, MediaPipe for 478-point landmark extraction, and dlib embeddings for accurate face identification.
- Designed modular architecture with a Qt6 GUI and a filesystem-based image storage layer with JSON indexing.
- Implemented Euclidean distance similarity search across stored 128-D embeddings enabling accurate face detection.

Custom USB Game Controller | ***(RP2040, C/C++, USB HID, CAD, 3D Printing)*** (September 2026 - Present)
- Designing a custom wired USB game controller using an RP2040 microcontroller, integrating Hall-effect joysticks, analog triggers, and digital button inputs for PC gaming.
- Developing embedded firmware using GPIO, onboard ADC, I²C, and USB HID to process six analog input channels while prototyping a 3D-printed enclosure and internal component layout for custom electronics, wiring, and assembly.

RC Car Control Interface | ***(Frontend Developer)*** | ***(React, TypeScript, Vite, Reset APIs)*** (April 2026 - June 2026)
- Developed a React and TypeScript frontend for an RC car telemetry chatbot, designing an interactive interface for querying vehicle data through natural-language inputs.
- Worked on frontend architecture for integrating RESET API communication with a Python/FastAPI backend, supporting retrieval of vehicle telemetry from a Neo4j knowledge graph.

LEADERSHIP AND EXPERIENCE

UF Trailblazers First-Year Leadership Program **Gainesville, FL @ University of Florida**
(Cohort Member) (2025 - 2026)
- Selected as 1 of 33 first-year engineering students for a competitive year-long leadership program; organized academic and social development activities fostering peer support, accountability, and cross-disciplinary collaboration.
- Expanded professional and community impact by engaging with 10+ industry professionals and alumni through mentorship and career workshops, and partnering with 10+ campus organizations to execute service initiatives benefiting the Gainesville community.

EmpowerED Learners & Alli Leadership Institute **Jersey City, NJ**
(Co-Founder, Web Developer & Civic Leadership Fellow) (2023 - 2026)
- Co-founded EmpowerED Learners, a nonprofit advancing special education awareness; reached 1,200+ students through community and Board of Education partnerships and developed a multilingual resource website to improve accessibility for immigrant families.
- Selected for a competitive 8-week Alli Leadership Institute fellowship, working alongside Jersey City mayoral candidate Mussab Ali on campaign strategy, grassroots outreach, and community initiatives; collaborated with local stakeholders to increase civic awareness and strengthen community participation across Jersey City.

ACTIVITIES

Indo-Caribbean Student Association (ICSA) *(Board Member & Historian)* **University of Florida**
- Planned cultural programming celebrating Indo-Caribbean heritage and building campus community..
- Documented events and supported initiatives increasing student engagement by 50% across social media and attendance.

ColorStack *(Active Member)*
- Member of a national community supporting Black and Latinx students in tech, participating in mentorship, networking, workshops, and programs focused on career development, technical growth, and industry readiness.

National Society of Black Engineers *(Active Member)* **NYU & University of Florida Chapters**
- Led fundraising efforts generating $2K+ and contributed to initiatives increasing active membership by 25%.

SKILLS & INTEREST

- **Technical:** Python, HTML, CSS, GitHub, CAD(Beginner), 3D Printing

- **Other:** Research Writing, Public Speaking, Leadership, Event Planning